第8章 耦合电路

8.1 含耦合电感的正弦电路

8.2 含耦合电感电路的计算

8.3 空心变压器电路的分析

8.4 理想变压器



4 有互感的电路的计算

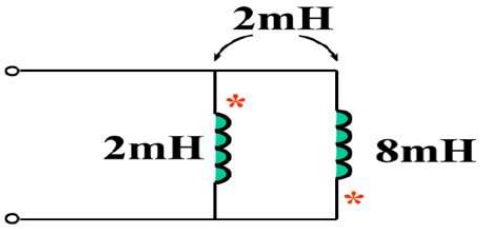
有互感的电路的计算仍属正弦稳态分析，前面介绍的相量分析的的方法均适用。只需注意互感线圈上的电压除自感电压外，还应包含互感电压。



雨课堂 Rain Classroom

单选题 1分

该端口的去耦等效电感为



A 0.857 mH

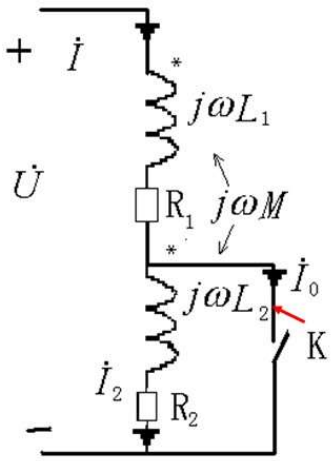
B 0.962 mH

C 4.857 mH

D 2 mH

2023年5月8日

例7.1：求开关K打开时和闭合时的电流



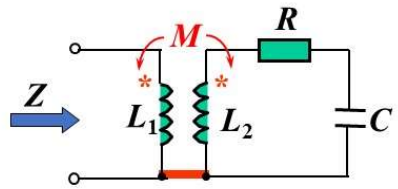
解：(1) K打开时

$$\dot{I} = \frac{\dot{U}}{R_1 + R_2 + j\omega(L_1 + L_2 + 2M)}$$

(2) K闭合时

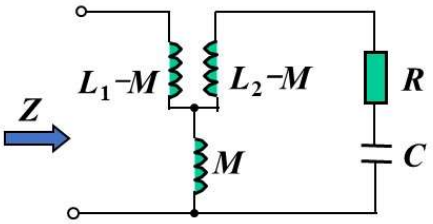
$$(R_1 + j\omega L_1)\dot{I} + j\omega M\dot{I}_2 = \dot{U}$$
$$j\omega M\dot{I} + (R_2 + j\omega L_2)\dot{I}_2 = 0$$

例1：已知如图，求入端阻抗 Z=?



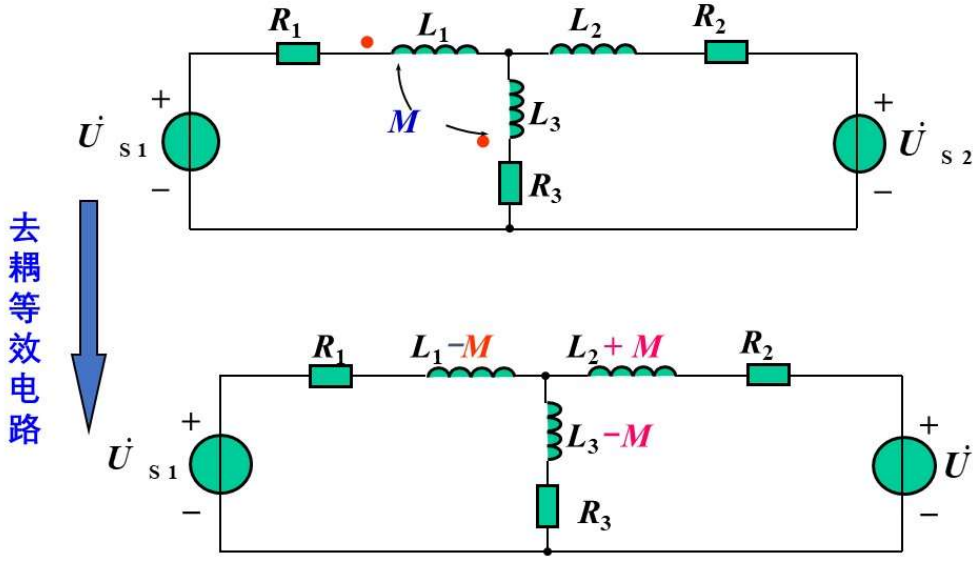
法一　端口加压求流

法二　去耦等效



2023年5月8日

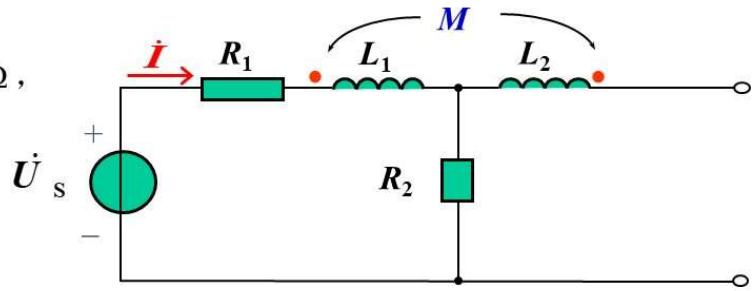
例2　画出下图电路的去耦等效电路。

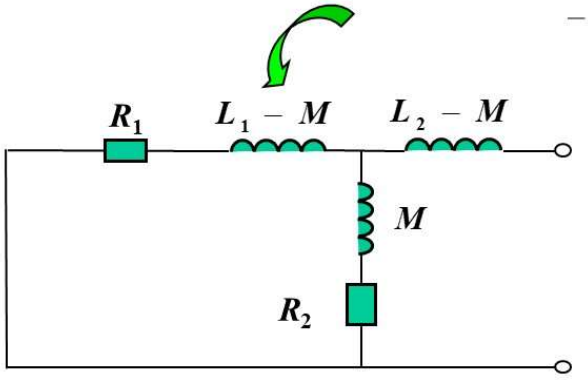


例3　求戴维南阻抗

$\omega L_1 = \omega L_2 = 10\Omega$，　$\omega M = 5\Omega$，

$R_1 = R_2 = 6\Omega$



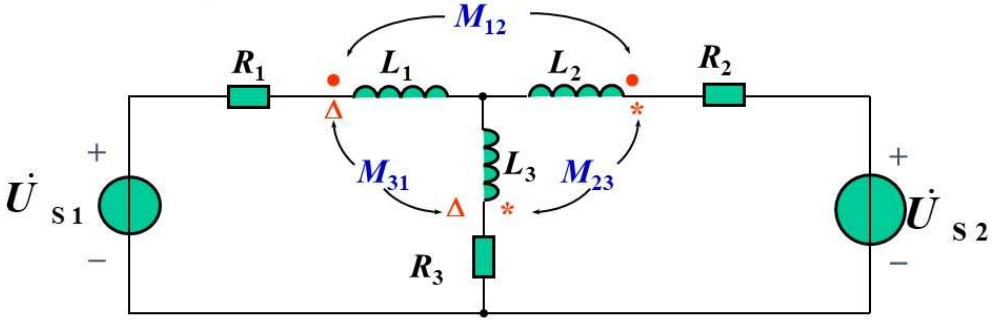


$$Z_i = j\omega(L_2 - M) + \frac{[R_1 + j\omega(L_1 - M)](R_2 + j\omega M)}{R_1 + j\omega(L_1 - M) + R_2 + j\omega M}$$

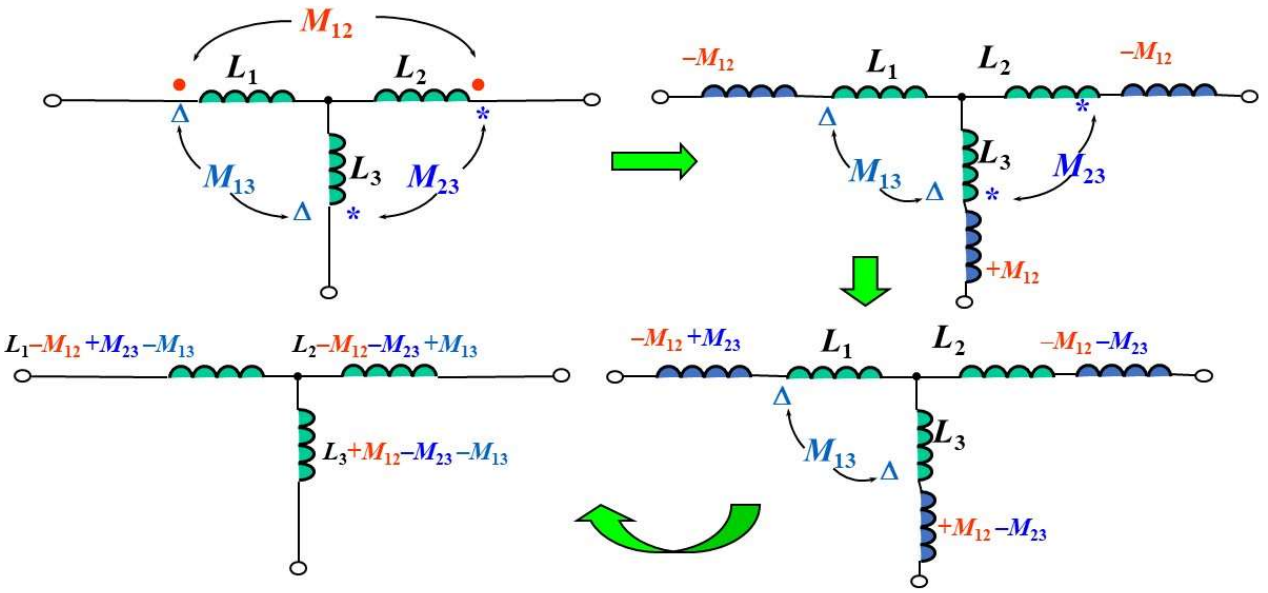
$$= j5 + \frac{(6 + j5)(6 + j5)}{(6 + j5) + (6 + j5)}$$

$$= 3 + j7.5\ \Omega$$

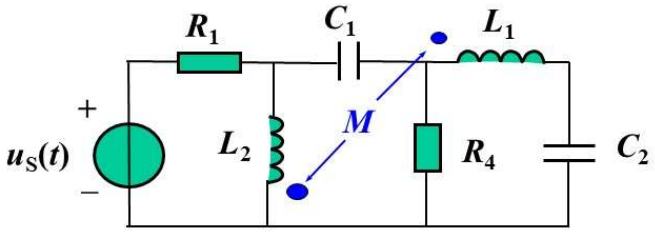
Ex: 画出去耦电路。



去耦等效电路(一对一对消)



去耦等效不是万能的



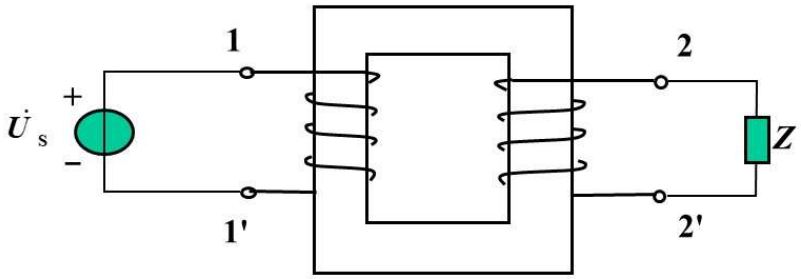
没有公共点

怎么办?



变压器铁心：硅钢片叠压而成。变压器绕组：高强度漆包线绕制而成。其它部件：油箱、冷却装置、保护装置等。

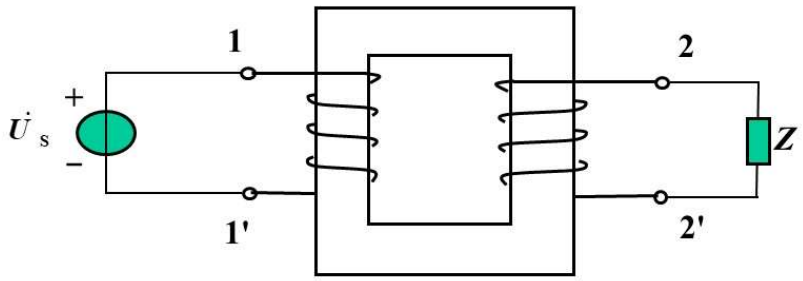
变压器 (Transformer)



利用互感的作用来传递能量

- 交流变压、变流
- 传送功率
- 电隔离
- 阻抗匹配

研究思路



1 考虑线圈内阻，求从原边(副边)看的等效电路

空芯变压器模型

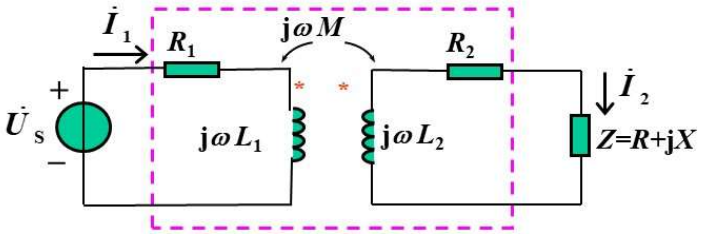
2 忽略线圈内阻，耦合系数为1

全耦合变压器模型

3 电感值趋向于无穷大

理想变压器模型

8.3 空心变压器



$$\begin{cases} Z_{11}\dot{I}_1 - j\omega M\dot{I}_2 = \dot{U}_S \\ -j\omega M\dot{I}_1 + Z_{22}\dot{I}_2 = 0 \end{cases}$$

$$\dot{I}_2 = \frac{j\omega M\dot{I}_1}{Z_{22}}$$

$$\dot{I}_1 = \frac{\dot{U}_S}{Z_{11} + \dfrac{(\omega M)^2}{Z_{22}}}$$

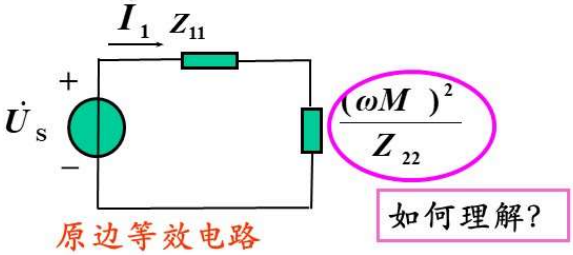
原边回路总抗阻

$Z_{11}=R_1+j\omega L_1$

副边回路总阻抗

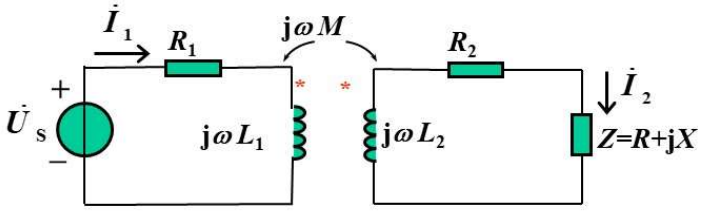
$Z_{22}=(R_2+R)+j(\omega L_2+X)$

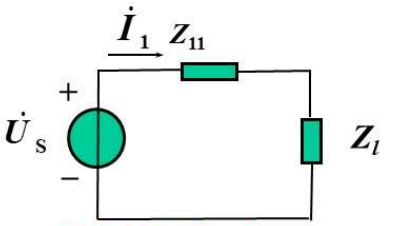
$$Z_{in} = \frac{\dot{U}_S}{\dot{I}_1} = Z_{11} + \frac{(\omega M)^2}{Z_{22}}$$



原边等效电路

如何理解?





原边等效电路

$$Z_{in}=\frac{\dot{U}_S}{\dot{I}_1}=Z_{11}+\frac{(\omega M)^2}{Z_{22}}$$

引入电阻　　引入电抗

$$Z_l \overset{def}{=} \frac{(\omega M)^2}{Z_{22}}=\frac{\omega^2 M^2}{R_{22}+jX_{22}}=\frac{\omega^2 M^2 R_{22}}{R_{22}^2+X_{22}^2}-j\frac{\omega^2 M^2 X_{22}}{R_{22}^2+X_{22}^2}=R_l+jX_l$$

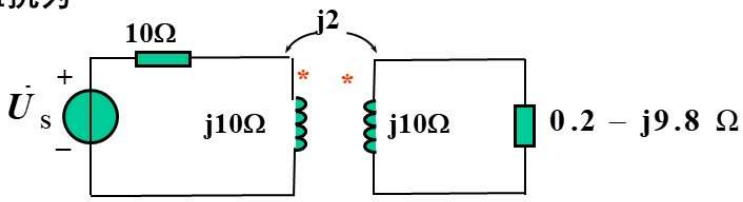
$Z_l=R_l+jX_l$　副边反映在原边回路中的阻抗（引入阻抗）。

当 $\dot{I}_2=0$，　即副边开路，$Z_{in}=Z_{11}$

当 $\dot{I}_2\neq 0$，　$Z_{in}=Z_{11}+Z_l$

单选题　1分

副边反映在原边回路中的引入阻抗为



A　$10+j10\ \Omega$

B　$8.32+j0.41\ \Omega$

C　$-j0.4\ \Omega$

D　$10-j10\ \Omega$

$Z_{11}=R_1+j\omega L_1$

$Z_{22}=(R_2+R)+j(\omega L_2+X)$

$$Z_l \overset{def}{=} \frac{(\omega M)^2}{Z_{22}}$$

能量角度分析

$$R_l = \frac{\omega^2 M^2 R_{22}}{R_{22}^2 + X_{22}^2} > 0$$

原边看：

电源发出有功＝电阻吸收有功＝$I_1^2(R_1+R_l)$

$I_1^2R_1$ 消耗在原边；

$I_1^2R_l$ ？

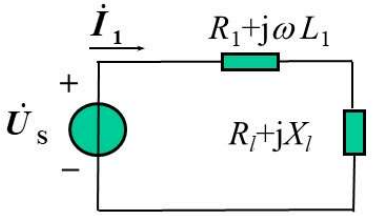
副边看：

$$\dot{I}_2 = \frac{\mathrm{j}\omega M \dot{I}_1}{Z_{22}}$$

副边吸收的有功：

$$I_2^2 R_{22} = I_1^2 \times \frac{\omega^2 M^2 R_{22}}{R_{22}^2 + X_{22}^2}$$

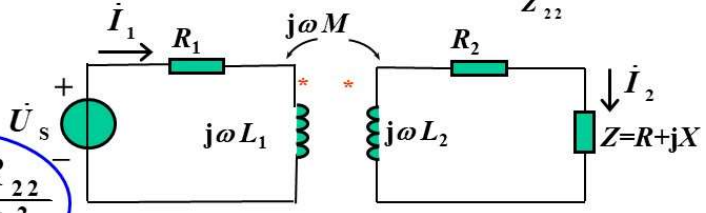
$$= I_1^2 R_l$$



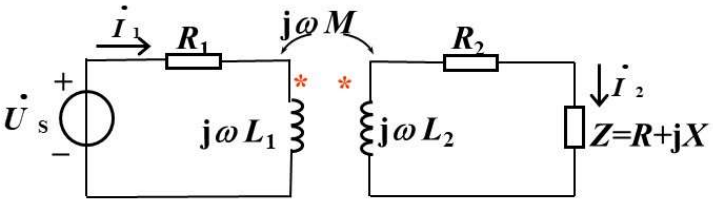
$Z_{11}=R_1+\mathrm{j}\omega L_1$

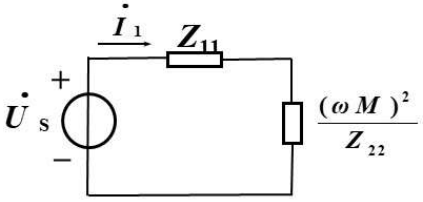
$Z_{22}=(R_2+R)+\mathrm{j}(\omega L_2+X)$

$$Z_l \overset{\text{def}}{=} \frac{(\omega M)^2}{Z_{22}}$$



变压器实现功率传送





原边等效电路

从能量角度来说：

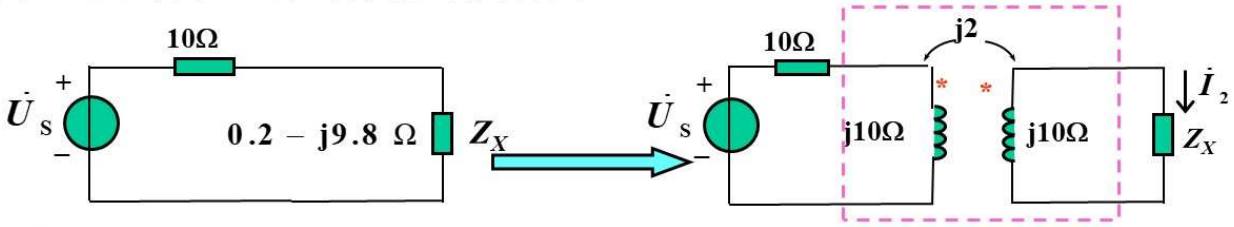
$$R_l = \frac{\omega^2 M^2 R_{22}}{R_{22}^2 + X_{22}^2} > 0$$

电源发出的有功＝电阻吸收的有功＝$I_1^2(R_1+R_l)$

$I_1^2R_1$ 消耗在原边；　　$I_1^2R_l$ 消耗在副边。

实现了功率的传送。

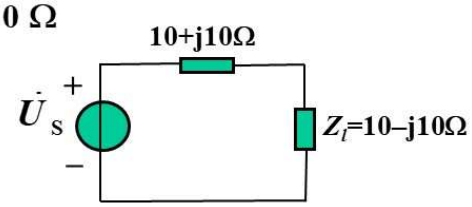
例4　已知：电压源U_S=20 V。 在电源和负载间加入变压器（如图），验证电路处于最佳匹配；并求此时负载获得的有功功率。



引入阻抗

$$Z_l = \frac{\omega^2 M^2}{Z_{22}} = \frac{4}{0.2 - j9.8 + j10} = 10 - j10\ \Omega$$

$$Z_l = Z_{11}^*$$ 最佳匹配

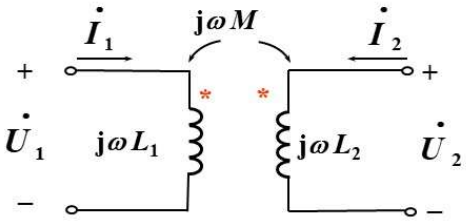


此时负载Z_X获得的有功功率

$$P = P_{R_{引入}} = \frac{20^2}{4 \times 10} = 10\ \text{W}$$

变压器实现共轭匹配

3 全耦合变压器 (*unity-coupled transformer*)



$$\begin{cases} \dot{U}_1 = j\omega L_1 \dot{I}_1 + j\omega M \dot{I}_2 \\ \dot{U}_2 = j\omega L_2 \dot{I}_2 + j\omega M \dot{I}_1 \end{cases}$$

$$\dot{I}_1 = \frac{\dot{U}_2 - j\omega L_2 \dot{I}_2}{j\omega M}$$

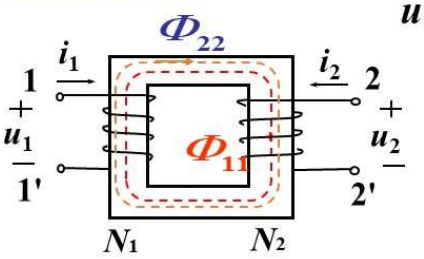
$$\dot{U}_1 = \frac{L_1}{M}(\dot{U}_2 - j\omega L_2 \dot{I}_2) + j\omega M \dot{I}_2 = \frac{L_1}{M}\dot{U}_2$$

全耦合时：　$M = \sqrt{L_1 L_2}$ ，$k = 1$

$$\frac{\dot{U}_1}{\dot{U}_2} = \frac{L_1}{M} = \frac{M}{L_2} = \sqrt{\frac{L_1}{L_2}} = \frac{N_1}{N_2} = n$$

*n*称为变比

从磁通分析:



$$u_1 = N_1 \frac{\mathrm{d}\Phi_1}{\mathrm{d}t}, \quad u_2 = N_2 \frac{\mathrm{d}\Phi_2}{\mathrm{d}t}$$

$$\Phi_1 = \Phi_2 = \Phi_{11} + \Phi_{22}$$

$$\frac{u_1}{u_2} = \frac{N_1}{N_2} = n$$

$$\dot{U}_1 = \mathrm{j}\omega L_1 \dot{I}_1 + \mathrm{j}\omega M \dot{I}_2$$

$$\Rightarrow \quad \dot{I}_1 = \frac{\dot{U}_1 - \mathrm{j}\omega M \dot{I}_2}{\mathrm{j}\omega L_1} = \frac{\dot{U}_1}{\mathrm{j}\omega L_1} - \frac{\mathrm{j}\omega M}{\mathrm{j}\omega L_1}\dot{I}_2 = \frac{\dot{U}_1}{\mathrm{j}\omega L_1} - \frac{1}{n}\dot{I}_2$$

全耦合变压器的电压、电流关系:

$$\begin{cases} \dot{U}_1 = n\dot{U}_2 \\ \dot{I}_1 = \dfrac{\dot{U}_1}{\mathrm{j}\omega L_1} - \dfrac{1}{n}\dot{I}_2 \end{cases}$$



8-4 理想变压器*ideal transformer*

理想变压器是一种特殊的无损耗全耦合变压器.

理想变压器的定义

同时满足下面三个充要条件

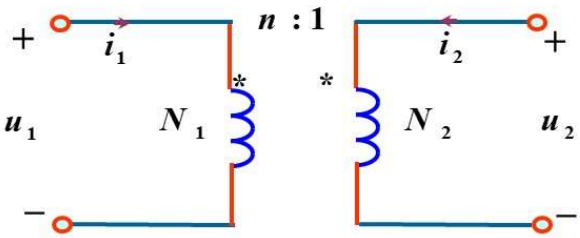
(1) 变压器本身无损耗;

(2) 全耦合K=1

满足 $\sqrt{\dfrac{L_1}{L_2}} = \dfrac{N_1}{N_2} = n$

(3) L_1, L_2, M均为无穷大.

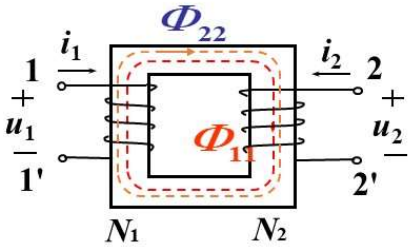
但 $\sqrt{\dfrac{L_1}{L_2}} = \dfrac{N_1}{N_2} = n$ 仍成立 .



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理想变压器



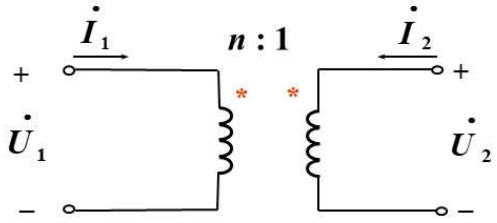
全耦合变压器

$$\begin{cases}\dot{U}_1 = n\dot{U}_2 \\ \dot{I}_1 = \dfrac{\dot{U}_1}{\mathrm{j}\omega L_1} - \dfrac{1}{n}\dot{I}_2\end{cases}$$

当$L_1, M, L_2 \to \infty$，L_1/L_2比值不变(磁导率$\mu \to \infty$)，则有

$$\begin{cases}\dot{U}_1 = n\dot{U}_2 \\ \dot{I}_1 = -\dfrac{1}{n}\dot{I}_2\end{cases}$$

理想变压器的元件特性

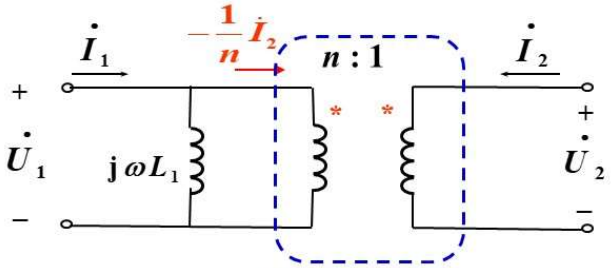


理想变压器的电路模型

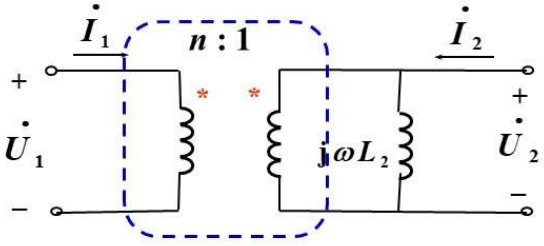


用理想变压器表示的全耦合变压器的电路模型:

$$\begin{cases}\dot{U}_1 = n\dot{U}_2 \\ \dot{I}_1 = \dfrac{\dot{U}_2 - \mathrm{j}\omega L_2\dot{I}_2}{\mathrm{j}\omega M} = \dfrac{1}{\mathrm{j}\omega Mn}\dot{U}_1 - \dfrac{\mathrm{j}\omega L_2}{\mathrm{j}\omega M}\dot{I}_2 = \dfrac{\dot{U}_1}{\mathrm{j}\omega L_1} - \dfrac{1}{n}\dot{I}_2\end{cases}$$



理想变压器



理想变压器

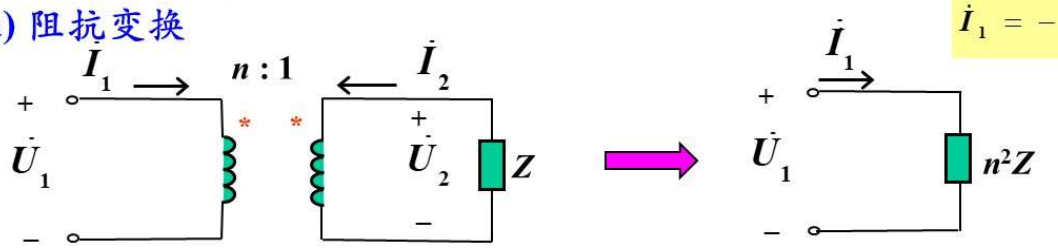


理想变压器的性质：

$\dot{U}_1 = n\dot{U}_2$

$\dot{I}_1 = -\frac{1}{n}\dot{I}_2$

(a) 阻抗变换



$$\frac{\dot{U}_1}{\dot{I}_1} = \frac{n\dot{U}_2}{-\frac{1}{n}\dot{I}_2} = n^2(-\frac{\dot{U}_2}{\dot{I}_2}) = n^2 Z$$

(b) 功率消耗

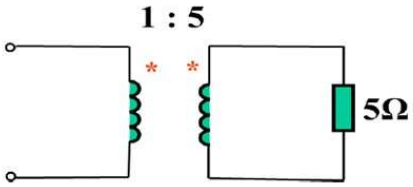
$$p = u_1 i_1 + u_2 i_2 = n u_2 \times (-\frac{1}{n}) i_2 + u_2 \times i_2 = 0$$

理想变压器既不储能，也不耗能，
在电路中只起传递信号和能量的作用。



单选题 1分

从理想变压器原边看进去，等效电阻为



A 0.2 Ω

B 1 Ω

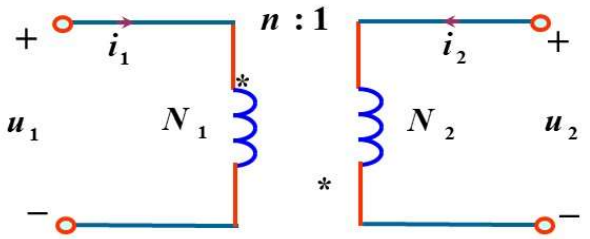
C 25 Ω

D 125 Ω

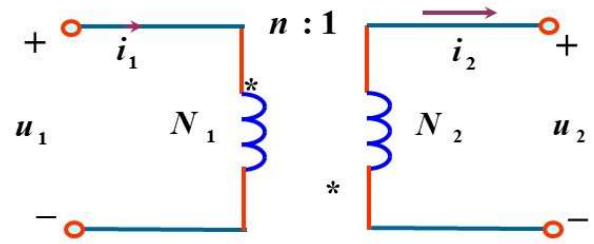


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电压电流关系如何?



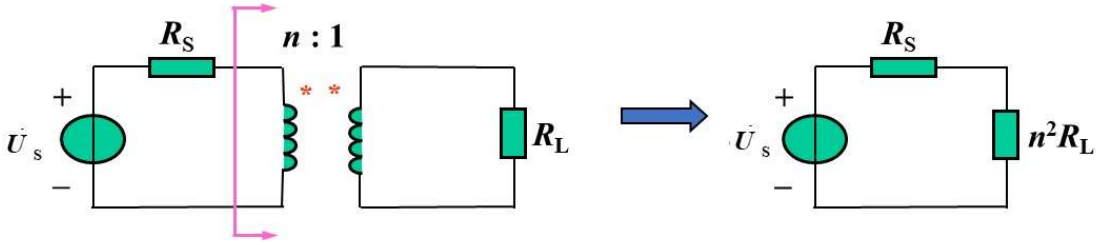
$$\begin{cases} \dot{U}_1 = -n\dot{U}_2 \\ \dot{I}_1 = \dfrac{1}{n}\dot{I}_2 \end{cases}$$



$$\begin{cases} \dot{U}_1 = -n\dot{U}_2 \\ \dot{I}_1 = -\dfrac{1}{n}\dot{I}_2 \end{cases}$$

2023年5月8日

例5 已知电阻R_S=1kΩ，负载电阻R_L=10Ω。为使R_L上获得最大功率，求理想变压器的变比n。

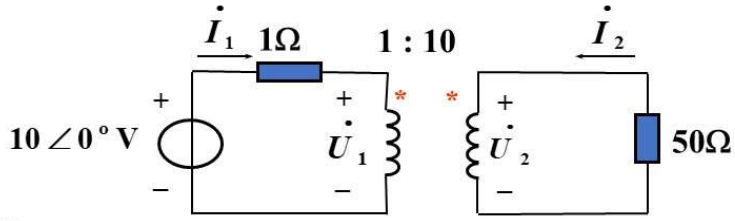


当 $n^2R_L=R_S$时匹配，即

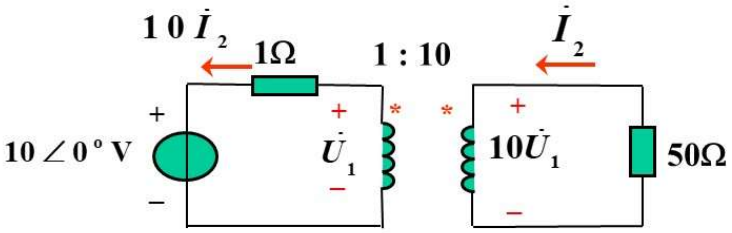
$10n^2=1000$

$n=10$

例6　求 $\dot{U}_2$.



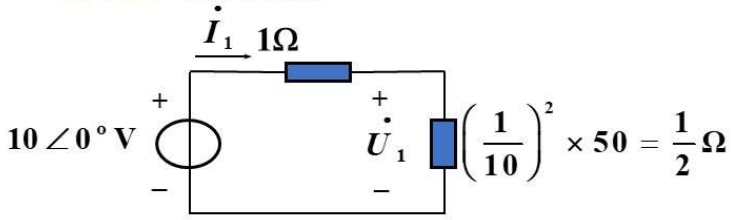
方法1：列方程



$$\begin{cases} 10\dot{U}_1 = -50\dot{I}_2 \\ 10\dot{I}_2 + 10 = \dot{U}_1 \end{cases}$$

$$\dot{U}_2 = 33.3\angle 0^\circ \text{ V}$$

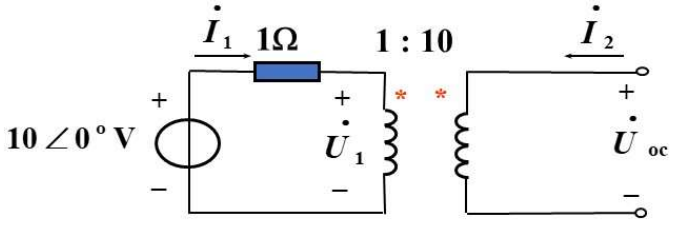
方法2：阻抗变换



$$\dot{U}_1 = \frac{10\angle 0^\circ}{1+1/2} \times \frac{1}{2} = \frac{10}{3}\angle 0^\circ \text{ V}$$

$$\dot{U}_2 = 10\ \dot{U}_1 = 33.33\angle 0^\circ \text{ V}$$

方法3：戴维南等效



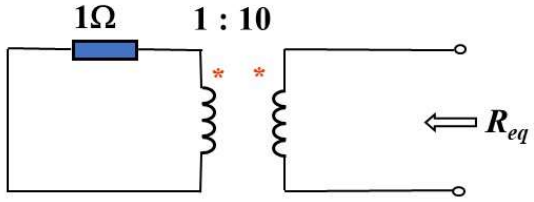
求 $\dot{U}_{oc}$ ：

$\because \dot{I}_2 = 0, \quad \therefore \dot{I}_1 = 0$

$$\dot{U}_{oc} = 10\ \dot{U}_1 = 10\ \dot{U}_S = 100\angle 0^\circ \text{ V}$$

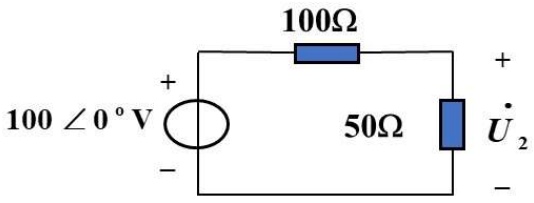
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求R_{eq}：



$R_{eq}=10^2\times1=100\Omega$

戴维南等效电路：



$$\dot{U}_2 = \frac{100\ \angle 0^\circ}{100\ +\ 50} \times 50\ =\ 33.33\ \angle 0^\circ\ \mathrm{V}$$